APURV DAS

Python Developer — Computer Vision (CNN Image Search) and Network Risk Modelling

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SUMMARY

MCA graduate who completed an IBM-partnered internship building an AI visual search engine for e-commerce (CNN feature extraction, image similarity, TensorFlow/Keras, OpenCV). Also built NetRisk AI, a home-network risk assessment tool with a per-device anomaly model that learns how often each port is open, running unattended on a 4 GB laptop. Writes tested Python (pytest, mocks) and ships with CI.

TECHNICAL SKILLS

- **Machine Learning / AI:** CNN, Feature Extraction, Image Similarity Matching, Model Training & Fine-Tuning, TensorFlow, Keras, OpenCV, pandas, Probabilistic Anomaly Detection
- **Languages:** C++20, Python, JavaScript, TypeScript, PHP, SQL, HLSL (Shader Model 4), HTML5, CSS3, Rust (basic, Tauri)
- **Databases:** PostgreSQL, Supabase (Row-Level Security, RPCs), MySQL, SQLite, IndexedDB
- **Test Tools:** Playwright, pytest, PHPUnit, Vitest, CTest, unittest, JIRA, Android Studio, Android Debug Bridge (ADB), Android Emulator, Chrome DevTools
- **DevOps & Tools:** Git, GitHub Actions (CI/CD), GitHub Pages, CMake, vcpkg, MSVC, PowerShell, Docker, AWS

WORK EXPERIENCE

QA Tester & Product Contributor (Part-Time) — Zentrax Solutions LLP | Apr 2026 – Present

- Perform manual functional, regression, exploratory and UI/UX testing of an intercity cab-sharing platform: the Android app, the web admin panel (backoffice) and its backend APIs.

AI / ML Intern (Remote Mentoring Program) — Pheme Software Pvt. Ltd. (IBM Platinum Business Partner), Remote | Jan 2026 – May 2026

- Built "AI Visual Search Engine for E-Commerce", an image-based product search system using CNN feature extraction and image similarity matching in Python.
- Trained and fine-tuned Convolutional Neural Network (CNN) models to improve visual search relevance and retrieval accuracy.
- Applied deep learning and computer vision (TensorFlow/Keras, OpenCV) to a real-world e-commerce use case, mentored by Mr. Krishna K.

Web Developer Intern — Technotask Business Solutions, Bhopal | May 2023 – Jun 2024

- Developed and maintained web applications using PHP, CodeIgniter (MVC), HTML5, CSS3, JavaScript and MySQL.

PROJECTS

NetRisk AI — Home-Network Risk Assessment — Course Project | Oct 2026

Tools: Python, nmap, SQLite, Windows Task Scheduler

- Finds every device on the home subnet and scans the top 200 TCP ports with versions plus a UDP probe for SNMP and UPnP, every 6 hours, unattended on a 4 GB laptop.
- Scores each finding by likelihood × impact into device and network risk (Low to Critical); an anomaly model learns how often each port is open on each device and flags scans far more surprising than that device's history.
- Keeps a SQLite history (new devices, newly opened ports), writes a self-contained HTML report with fix steps, and sends a Windows notification only for High/Critical or unusual findings.
- Safe by design: alerts only, never changes the network, pins the home network on first run and skips any other network; 44 unit tests with mocks.

Project Manager — Offline-First Web App + Windows Desktop App — Personal Project | Aug 2026 – Present

Tools: Supabase/PostgreSQL, JavaScript, Tauri (Rust), Playwright, GitHub Actions | Code: github.com/ApurvDas/project-manager

- Live Supabase app with an offline-first sync engine (IndexedDB plus an outbox replayed through server RPCs) and a Tauri Windows desktop app using about 170 MB idle against about 315 MB for Chrome; 60 Playwright E2E tests and 32 API permission tests.

Other Projects — Personal | 2026

- Unigen Manager (Python, pytest) and Focus (FastAPI, pytest, GitHub Actions CI): each with automated tests.

EDUCATION

Master of Computer Applications (MCA) — UPES, Dehradun | Aug 2024 – Jul 2026 (Completed)

Bachelor of Computer Applications (BCA) — Christ (Deemed to be University), Delhi NCR | Apr 2024

Higher Secondary — NIOS | Jun 2020